Alternative species concepts

Although Mayr's idea of biological speciation is perhaps the most common way to define species and explain how they evolve, it is far from the only one. In fact, there are more than 20 recognized species concepts, ranging across two broad groups: typological and evolutionary concepts.

Typological species concepts are based on the idea that a population of individuals of the same type—or sharing the same set of traits—are what makes up a species. The traits can be based on genetics, such as DNA or RNA base sequences, or on phenotypes, such as the size of certain body parts or particular markings, such as the arrangements of spots on insects' wings. The evolutionary species concept is based on species lineages. A species is defined as the organisms that share a lineage from the time when the species initially split off until extinction, or until an additional splitting off and creation of a new species.